

Two Modes, Not Many Types: The Structure of Football Player Roles in Season-Aggregate Data

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Abstract

Football analysts sort players into role archetypes such as the ball-playing centre back or the deep-lying playmaker. Whether such types exist has never been tested, because a clustering algorithm returns groups whatever it is given, so the usual evidence for a taxonomy is satisfied just as well by a smooth continuum. We test existence directly on 13,822 player-seasons from five seasons of the big five European leagues, described by 33 statistics, by calibrating cluster quality against simulated data matched to the sample in size, dimension and correlation but built to contain no clusters. One division survives, and it is coarse: players separate into a defensive and an attacking mode of involvement, exceeding every simulation under all three clusterless references in every scope, by at least 8.01 standard deviations against the hardest. Nothing finer survives, and the usual stability evidence cannot make the distinction, since clusterless data resample at an adjusted Rand index of at least 0.911. The two modes are not the listed positions renamed, agreeing with them at 0.293, and they reproduce when each season and each league is fitted independently. A club’s composition along the axis predicts league points beyond possession out of sample, while the finer archetypes predict nothing further. Football role space has two poles and a populated middle, and finer taxonomies are artefacts of the method used to find them.

Keywords: cluster validity; clusterability; null calibration; soccer analytics; unsupervised learning; reproducible research

1 Introduction

Public football data now describes every player in a major league by dozens of per-match measures, and a standard genre of analysis has grown around it. Take the measures, group the players by how similar their measures are, name the groups, and present the result as a taxonomy of roles. Published examples report four archetypes within a single position group, and in basketball nine or thirteen across the whole sport, with labels such as deep-lying progressor or advanced playmaker [1–3]. The appeal is obvious. A taxonomy turns a table of numbers into a vocabulary a coach or a recruiter can use.

The grouping step is done by a clustering algorithm, and it is worth being explicit about what one does, because the whole argument of this paper turns on it. A clustering algorithm is handed a set of points, here one point per player with one coordinate per statistic, and a number of groups to find. It then divides the points into that many groups so as to make each group as internally similar

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as possible. Nothing in that procedure asks whether the points were grouped to begin with. Handed a single cloud of points with no groups in it, a clustering algorithm still divides the cloud, and the division it returns will still look tidy by the measures normally used to judge one. Running the standard method on a single multivariate Gaussian, which by construction contains no groups at all, returns two clean clusters with a respectable separation score. Getting clusters out of a clustering algorithm is therefore not evidence that clusters were there to begin with.

The football literature has not tested that inference, mostly because it has been busy with a harder data problem. The most detailed work on roles uses proprietary feeds: formations and positional roles recovered from tracking coordinates, player vectors and performance ratings learned from event streams, and playing styles distinguished from match events [4–7]. Where public season aggregates are used instead, the standard output is a taxonomy, whether from k -means on a dozen indicators or from partial membership models on a season of counts [1, 8]. Two systematic reviews catalogue the genre for players and for teams, the first synthesising twenty-six styles from twelve studies and finding no shared framework behind them [9, 10]. A review of 278 sports science articles that used clustering found that 55 percent did not state the criterion by which the number of clusters was chosen, which puts the omission in context [11]. The question of existence has not been answered badly so much as not asked.

Outside sport it has been asked for forty years, and the tools already exist. Choosing the number of clusters has been studied systematically since thirty stopping rules were compared on data with known structure, and the silhouette and the gap statistic descend from that programme [12–14]. The gap statistic is the direct ancestor of what this paper does, since it compares the observed clustering against the same quantity computed on reference data drawn uniformly over the principal box. A second tradition validates a partition by its stability under resampling, and the theory reached a conclusion the applied literature has largely not absorbed: for algorithms such as k -means, stability is governed by the geometry of the objective rather than by the presence of clusters, so a partition can be perfectly stable on data with no groups in it [15, 16]. A third asks whether data are clusterable at all, finds the formal notions of it mutually inconsistent, and recommends tests built on Hartigan’s dip statistic applied to a low dimensional projection [17–19]. Underlying all three is the position that there is no true clustering independent of the purpose a clustering is meant to serve [20].

The closest methodological relative of this paper is the aggregated validity index of Akhanli and Hennig, which calibrates several criteria against their distribution over random partitions of the same data and, applied to a season of player statistics from eight European leagues, returned a coarse partition into five ordered groups [21, 22]. That coarse partition runs along the same territorial axis reported here. The difference is what the calibration is against. Comparing against random partitions of the observed points asks how good a clustering is relative to other clusterings of the same data, which is the right question for choosing among methods. Comparing against clusterless data matched to the observed sample asks whether any clustering of these points is better than a clustering of points with no structure in them, which is the question of existence. The two are complementary, and the first does not answer the second. A different tradition sidesteps the question by describing athletes as mixtures rather than as members, either as convex combinations of a few extreme profiles or as partial members of several clusters at once [8, 23]. Both are natural descriptions of a continuum and both presuppose one, since neither asks whether the data would have supported discrete groups instead.

This paper supplies the missing test. We assemble 13,822 player-seasons covering five seasons across the five major European leagues, with 33 per-90 features spanning shooting, chance creation, ball progression, territory, take-ons and defensive work. On that sample we run the standard pipeline, a selection rule fixed before any fitting followed by k -means, and then calibrate what it returns against simulated data built to contain no clusters at all. Three such references are used rather than

one, ranging from a Gaussian matched on covariance to a copula that keeps every real marginal distribution and correlation, so that a reader can see which reference the claim survives. The same simulations are used to measure what a resampling stability check is worth, which turns an assertion about the genre’s second standard piece of evidence into a number.

Our expectation, and what we find, is that the data support a real division but a far coarser one than the field reports. A single split survives calibration in every scope, separating players by defensive against attacking involvement, while every finer division is indistinguishable from what structureless data produces. That split is not the listed positions renamed, since defenders and forwards fall at opposite poles while midfielders divide almost evenly between them; it reproduces when each season and each league is fitted independently; and a club’s composition along it predicts league points beyond possession out of sample, while the finer archetypes predict nothing further. The practical implication is not that role analysis is impossible. It is that the correct output of role analysis on this kind of data is a position along an axis rather than membership of a type, and that a published taxonomy of a dozen archetypes is reporting structure its data cannot support.

2 Data and methods

Every procedure is seeded from a single constant, every hyperparameter is declared in one configuration file the analysis code reads, and nothing is tuned after seeing a result. Computation uses scikit-learn unless stated otherwise [24]. Procedural detail a reader does not need in order to follow the argument is in Section S1 of the supplement.

2.1 Sample

The analysis uses season-aggregate player statistics for the five major European leagues over five seasons from 2018 to 2022, labelled by the calendar year in which each season ends. It contains 13,822 player-seasons, of which 9,263 are outfield players above the eligibility threshold of 450 minutes and 1,035 are goalkeepers.

The tables originate from FBref, which published advanced statistics derived from a commercial event-data provider [25]. They are taken from a public mirror maintained by the worldfootballR project rather than from the site directly, and that needs stating plainly [26]. On 20 January 2026 the site’s operator removed every statistic derived from that provider, retroactively and across completed seasons, after the licence was terminated. The mirror predates the removal, so it is a secondary source in the strict sense: a third party scraped it, we did not observe the scrape, and it cannot be re-derived because the primary source no longer holds the data. What we can do, and do, is audit it, with row counts and per-column and per-season coverage checked on load as part of the pipeline. One season is excluded on those grounds, because the site restructured its ball-carrying columns for the season ending 2023 and the mirror did not follow the renames, leaving every carrying column empty while its neighbours are intact.

The outfield feature set contains 33 per-90 features spanning six families: shot volume and quality including non-penalty expected goals [27, 28]; chance creation; ball progression; touches across the five pitch zones; take-ons; and defensive work with passing accuracy by distance. Table S1 lists them with their source tables. Minutes and appearances are kept as context and never enter the feature set, because squad usage describes a player’s opportunity rather than the role he performs within it.

Defensive actions can only occur while the opponent has the ball, so their raw rates confound what a player does with how often his team is out of possession, which tracks team strength. The conventional correction assumes defensive volume is proportional to time spent out of possession [29]. That is an empirical claim, and we measure it rather than assume it: regressing log team

defensive volume per ninety on log opponent possession over 490 team-seasons gives an elasticity per statistic, and each count is adjusted by its own fitted value. Attacking statistics are deliberately left unadjusted, since they scale with a team’s own possession rather than the opponent’s. Section S3 reports the elasticities with bootstrap intervals and per-league refits, and tests that asymmetry rather than asserting it.

A second sample is pulled live for the seasons ending 2025 and 2026, after the withdrawal, with expected goals from Understat as an independent source [30, 31]. It retains 18 features, roughly half the primary set, and serves only as a robustness case, because a result holding on both a rich and an impoverished description of the same game is worth more than one holding on either alone. It also carries a warning that applies beyond this study: the affected tables are still served with complete column headers and one row per player, and only the values are gone, so a feature set assembled from the published schema resolves correctly and yields an empty matrix. Selecting features against observed values rather than documentation is now a necessary step with this source.

Season aggregates discard context, and that bounds every claim below. A tackle count does not say where on the pitch the tackles happened, and an expected goals total does not distinguish six poor chances from two good ones. The sample is recent European club football, and the clusters are descriptive: a label records how a player’s season looked, not why.

2.2 Clustering and the selection rule

Clusters are estimated with k -means, alongside a Gaussian mixture model whose Bayesian information criterion is reported over the same range and HDBSCAN, which is permitted to answer that nothing is clustered and does so by assigning points to noise [32]. Clustering runs globally and within each of the three outfield position groups, which are treated separately because correlations between features reverse sign across groups, so one pooled geometry would average over relationships pointing in opposite directions.

The number of clusters is chosen by a rule fixed before any fitting, which matters because four criteria applied after the fact can be made to justify almost any choice. Over k from 2 to 12 we compute the silhouette, the Davies-Bouldin index, the Calinski-Harabasz index and the gap statistic [13, 14]. The rule selects the smallest k whose silhouette lies within 0.02 of the best observed and which ranks in the top three on at least two of the silhouette, Calinski-Harabasz and gap criteria, with Davies-Bouldin breaking ties. Which criteria agreed is reported alongside the choice.

2.3 Testing whether clusters exist at all

Selecting a number of clusters presupposes that clusters are there to be counted, and that presupposition is the weak point of the genre. The criteria above are computed on the partition rather than against any alternative, so a silhouette of 0.273 sounds like evidence until one asks what silhouette structureless data of the same shape would produce.

Let $X \in \mathbb{R}^{n \times d}$ be the standardised feature matrix for a scope, $\pi_k(X)$ the partition k -means returns under a fixed seed, and $s_k(X)$ its mean silhouette. The number of clusters reported, $k^* = 2$, is not chosen by inspecting $s_k(X)$; it is returned by the rule declared above, and the calibration then asks one question about it.

Three generators produce clusterless data matched to X . The Gaussian generator draws from $\mathcal{N}(\hat{\mu}, \hat{\Sigma})$, matching size, dimension and correlation while containing no groups by construction. The copula generator reproduces every marginal distribution of X exactly, skew and bounds included, along with the observed rank correlations, destroying only the joint structure beyond them. The uniform generator fills the bounding box of X after rotation to its principal axes, which is the

reference distribution of the gap statistic [14]. A single Gaussian is the easiest of the three to beat, because skew alone raises the silhouette a partition can reach, so whichever is hardest in a given scope is treated as the calibration that counts. Section S1 states each generator formally.

For each generator g and each of $B = 25$ draws $X^{(g,b)}$, the identical pipeline computes $s_{k^*}(X^{(g,b)})$, giving a null sample of size B . The reported statistic is the standard score

$$z_g = \frac{s_{k^*}(X) - \bar{s}_g}{\hat{\sigma}_g},$$

taken against the mean and standard deviation of that null sample. With B draws the null standard deviation is itself an estimate, so z_g describes the margin rather than serving as a test statistic with a known distribution. The claim the paper makes is the distribution-free one, that $s_{k^*}(X)$ exceeds every one of the B null values under every generator, which under exchangeability between the observed and simulated statistics has Monte Carlo probability at most $1/(B+1)$ per generator and scope. The headline is a conjunction over three generators and four scopes rather than a disjunction, so it can only be weakened by adding a generator or a scope, never strengthened.

The same generators supply two further calibrations. Hartigan’s dip statistic on the leading principal component of each draw gives a null sample against which the observed dip is placed the same way, which matters because at these sample sizes the dip test’s p-value rejects unimodality for departures too small to be of interest [19]. The bootstrap adjusted Rand index of π_{k^*} is computed on the simulated datasets by exactly the resampling procedure applied to X [33], so the stability the genre reports for real data can be read against the stability a clusterless continuum produces. Rather than assert that a partition imposed on structureless data passes a bootstrap, the paper measures it.

One assumption should be stated because the analysis makes it silently otherwise. The rows of X are player-seasons and the same player appears in up to five of them, so rows are not independent. The generators match the marginal and pairwise structure of X but not this nesting, and the bootstrap resamples rows as if exchangeable. The replication below, which refits every season and every league independently, does not share this assumption, and it is why the paper rests no claim on a single pooled fit.

2.4 Validation

Five checks are applied to whatever the calibration accepts, each set out in full in Section S1. Stability comes from two hundred bootstrap resamples scored against the reference partition. Replication is tested more stringently, since a resample reuses the same players: each season and each league is standardised inside itself and clustered from scratch, borrowing no centre, no scale and no starting point, and one scope’s centroids are also imported into another scope’s players, which separates replication of the solution from change in the players. Robustness to the description is tested by grouping the features into six families, removing each in turn and trying each alone, and rerunning the selection rule and the calibration every time. Agreement with the listed positions is quantified by the adjusted Rand index and normalised mutual information, and independently by asking a gradient boosted classifier to predict position group from the same features [34, 35], where two properties of the confusion matrix are read as evidence about geometry rather than accuracy: whether the two extreme classes are confused with each other, and whether errors concentrate in the class between them. Finally the structure is tested against what happened, by summarising each team-season by its minutes-weighted composition and predicting points per match out of sample under leave-one-season-out, leave-one-league-out and leave-one-club-out holdouts against possession as the baseline. The increment over possession is the quantity of interest, because composition and

dominance are confounded and only the increment says whether the modes carry information that dominance does not.

Goalkeepers share almost no features with outfield players and are analysed separately on the 13 canonical goalkeeper features, under the same threshold, standardisation and selection rule, with saves above expectation computed as post-shot expected goals minus goals allowed so that shot difficulty is conditioned on rather than shot volume alone [36]. One choice is declared in advance because the result turns on it. The goalkeeper block mixes what a goalkeeper does with what his defence permits him to face, and a partition fitted on both will separate teams rather than goalkeepers, so we fit twice, once on the full block and once on technique alone, and report for each how much variance membership explains in situational against technical quantities.

2.5 Reproducibility

The analysis runs from a single command against pinned dependencies and a fixed seed. Every figure and every number in this paper and its supplement is generated by code in the accompanying repository and read into the document from generated definitions at build time, so no value was transcribed by hand. The code, the retrieval step that fetches the archive from its stated upstream with a recorded digest per file, the cluster assignments for both samples and the sources of both documents are at <https://github.com/ZacharyTChung/pl-player-profiling>.

3 Results

3.1 Is there any structure at all?

The first question is not how many clusters there are but whether there are any. Figure 1 answers it against the three clusterless references.

Real structure exists, and it is confined to a single division. At two clusters the observed silhouette is 0.273 for the outfield pool against a Gaussian null mean of 0.207, and it exceeds every one of the 25 simulations; the same holds in each position group. A Gaussian is the softest reference available, so the copula null, which keeps every feature’s real marginal distribution and rank correlations and so removes the objection that skew alone produced the separation, is the one to read. It is the hardest of the three in every scope, and the observed partition clears it everywhere, by 13.87 standard deviations on the pool and by 8.60, 10.13 and 8.01 within the three position groups. The observed silhouette at two clusters exceeds every simulation under every null in every scope, and the smallest margin anywhere, over all four scopes and all three nulls, is 8.01 standard deviations.

Beyond two clusters the curves converge, which is the central negative result stated precisely. By around six clusters the observed silhouette lies inside the simulated band in every position group, so the third and subsequent clusters a taxonomy would name are not distinguishable from what a partitioning algorithm extracts from structureless data.

The second standard piece of evidence for a published taxonomy is a bootstrap, and the same simulations show what it is worth. Computed on clusterless datasets by exactly the procedure applied to the real data, the bootstrap adjusted Rand index of the two-cluster partition never falls below 0.911 on any clusterless dataset in any scope, against 0.962 to 0.990 on the real data. The gap is as small as 0.003, and a stability screen at 0.8, demanding by the genre’s standards, is passed by every clusterless dataset in every scope. A resampling check does not distinguish real structure from a partitioned continuum: it measures whether an algorithm is deterministic on similar inputs, which k -means is.

Three further instruments agree, with one exception reported as such. The observed dip statistic

on the leading principal component sits 13.67 null standard deviations above the hardest reference for the pool, 8.84 for defenders and 5.66 for midfielders, so the dominant axis is two-moded rather than merely wide. Forwards are the exception: the dip test returns $p = 0.9936$ and the observed dip sits -2.69 standard deviations against the uniform reference, which is to say below it. The forward split is real by the silhouette calibration, clearing the hardest null by 8.01, and its leading axis is not bimodal by the dip. Both halves are reported because together they say what the forward split is, a cut through a continuum of involvement that the null cannot produce rather than two kinds of forward. Density-based clustering assigns 86.5 percent of the pool to noise and every position group entirely to noise, which is what a density method does when handed a continuum with a density gradient instead of separated groups, and the mixture criterion declines monotonically to the top of the candidate range in every scope, the signature of a model absorbing extra components into a smooth density.

Football role data contains a real division into two modes. It does not contain a taxonomy.

3.2 The shape of the space, and what the modes are not

A two-mode description invites two objections. If the data reduce to two groups, perhaps they reduce to one number; and if not, perhaps the two groups are the listed positions under another name. Figure 2 answers both, and the answer to each is no.

The leading principal component carries 39.9 percent of the variance, far more than any other single direction, which confirms that one axis dominates. It is nowhere near the whole space: the first three components together reach only 64.2 percent and 14 are needed to reach 90 percent of the variance, so the role space has one dominant axis and a long tail of smaller ones, which is a different object from a line. The content matches the football. The first component separates players by where on the pitch they touch the ball, loading positively on progressive receptions and touches in the attacking third and the opposition box and negatively on touches in the defensive third and in the defending team's own box. The second is a progression axis close to orthogonal to it, so a deep midfielder who moves the ball forward and a striker who receives it in the box sit at opposite ends of the second while sharing territory on the first. The first component is bimodal and drives the partition; the components behind it carry variation that is real, interpretable and continuous.

Projecting every player onto the axis between the two centroids shows the distribution the paper is describing: density concentrated at two ends with a populated middle, and the listed positions laid across that shape rather than aligned with it. Agreement with the listed position group is well above chance and well below recovery, at an adjusted Rand index of 0.293 and a normalised mutual information of 0.338, where a relabelling of position would return values near one and an unrelated partition values near zero. Where the line falls is the informative part. 92.9 percent of defenders fall on one side and 99.8 percent of forwards on the other, while midfielders divide almost evenly, 51.0 percent of them on the attacking side, and it is that even split which drags the agreement indices down. The partition is not a relabelling of position but a cut through it: the two extremes are assigned almost deterministically and the group between them is assigned by which end it leans toward.

A supervised model recovers the same geometry from machinery that shares nothing with the clustering. Gradient boosting predicts the listed position group at 93.4 percent accuracy out of fold against a majority-class baseline of 40.9 percent, with multinomial logistic regression barely behind at 93.1 percent; the near equality of a linear model and a boosted ensemble is itself informative, since a taxonomy of many discrete types would leave more interaction structure for the trees to find. The errors are the argument. Over 9,263 player-seasons, only 8 confusions link the two extreme classes directly, against 603 that run through midfield, so 1.31 percent of misclassifications skip

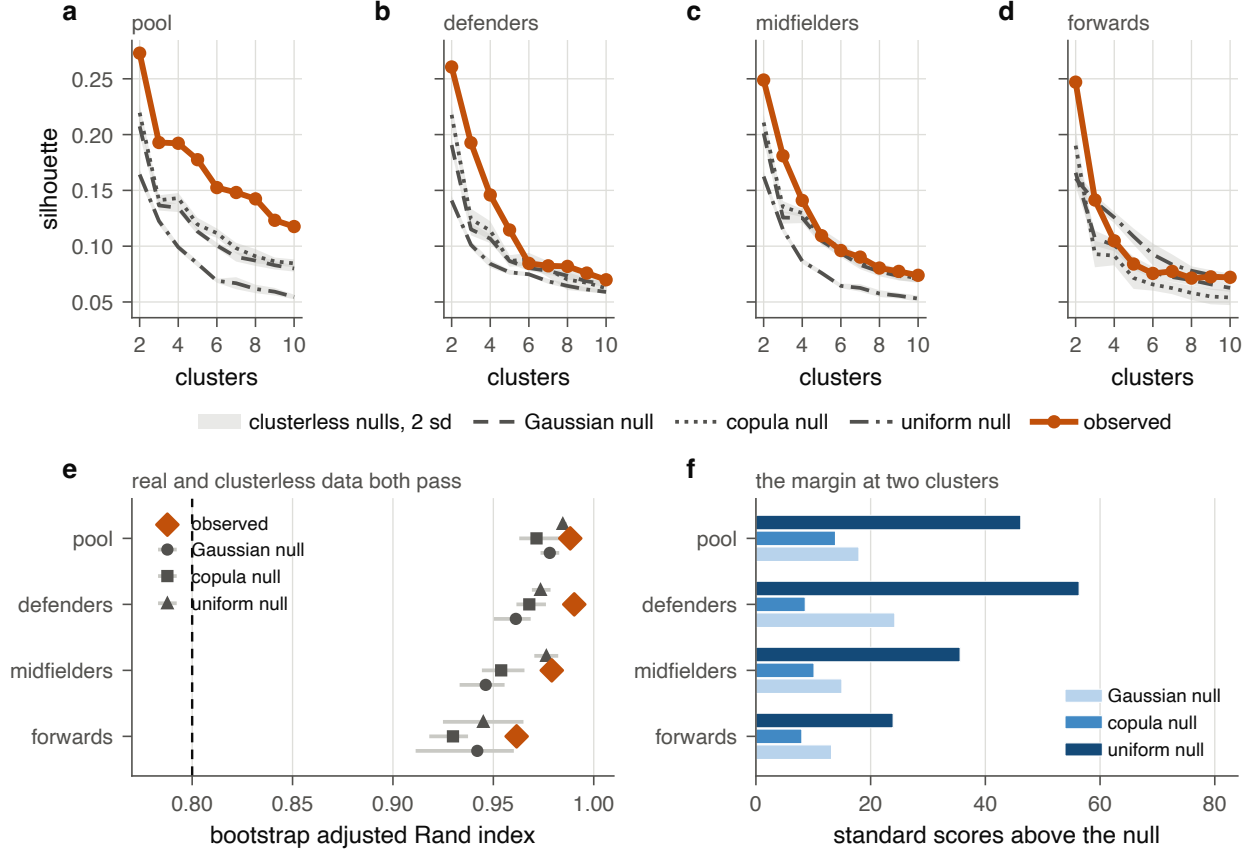


Figure 1: Two modes are real and nothing finer is. **a** to **d**, observed silhouette against three kinds of data built to contain no clusters, one panel per scope. Each null is simulated 25 times and drawn as its mean with a band of two standard deviations: a Gaussian matched on covariance, a copula that keeps every real marginal and correlation, and the uniform reference of the gap statistic. The observed curve separates from all three at two clusters and rejoins the bands as the number of clusters grows. **e**, bootstrap adjusted Rand index on the real data against the same quantity on the clusterless simulations, with the range over simulated datasets and the dashed line at 0.8, the screen a published taxonomy would cite as evidence that its partition is stable. The real data score a little higher than the clusterless data, by as little as 0.003, and both sides clear the screen comfortably, which is why passing it is not evidence that groups exist. **f**, the margin at two clusters, as standard scores above each null.

the middle. That is the confusion structure of an ordered line whose ends are far apart and whose middle overlaps both, not of three distinct categories. The attributions agree: the three strongest features overall are Touches def third, Clearances (adj) and Ball recoveries, all territorial rather than technical. Position, as the data record it, is mostly a statement about where a player stands.

3.3 What the two modes are

Two modes within each of three position groups give six descriptions. Names were assigned only after inspecting centroid profiles and member players, and the justifications are generated from the computed centroids rather than composed for the paper. Figure 3 shows the features that separate each pair.

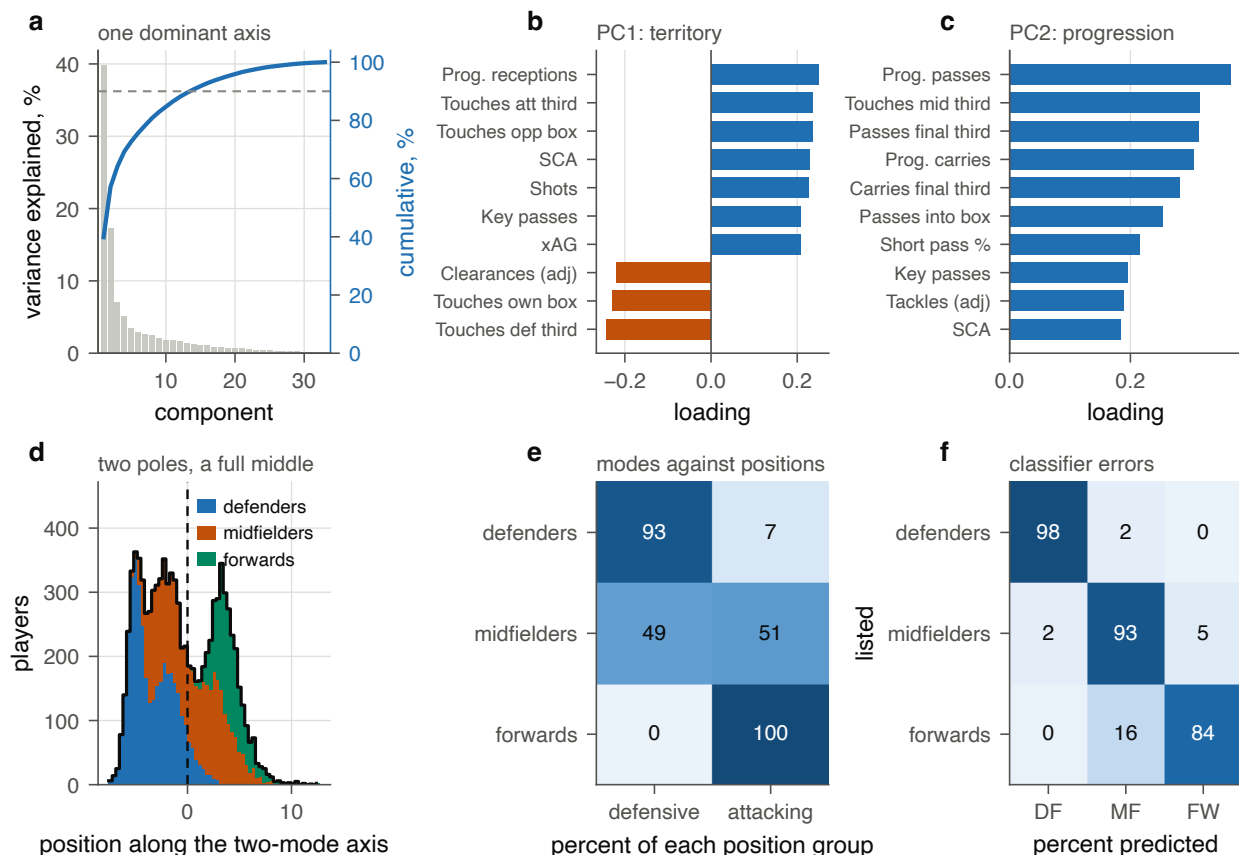


Figure 2: The space has one dominant axis and many smaller ones, and that axis is not the listed positions. **a**, variance explained per principal component with the cumulative curve, over 9,263 eligible player-seasons on 33 features. **b** and **c**, the ten strongest loadings on the first two components, which order players by pitch territory and by ball progression respectively. **d**, every player projected onto the axis between the two centroids, stacked by listed position group with the total drawn as an outline; the dashed line is the cut the partition makes. **e**, the share of each listed position group falling in each mode. **f**, out-of-fold confusion for the gradient boosted classifier, in percent of each listed class. The corners linking defenders and forwards are close to empty in both, and midfield is where the disagreement lives.

Among defenders the division is between 2,094 Deep central defenders and 1,699 Wide attacking defenders. The first take three times as many touches in their own defensive third, more than twice as many in their own box, twice the possession-adjusted clearances and win considerably more aerial duels; the second take five times as many touches in the attacking third, attempt ten times as many crosses and four times as many take-ons, and receive far more progressive passes. This is the centre back against wide defender distinction, and the notable point is that the source's own position label does not encode it, since the great majority of both groups are listed simply as defenders.

Among midfielders the division separates 1,939 Deep ball winning midfielders from 1,827 Advanced attacking midfielders. The first recover the ball more often, tackle more and take far more touches in their own third. The instructive detail is that they also make more passes into the final third and more progressive passes than the second group: they are not less involved in progressing the ball, they progress it differently, by passing forward from deep rather than by receiving high. A

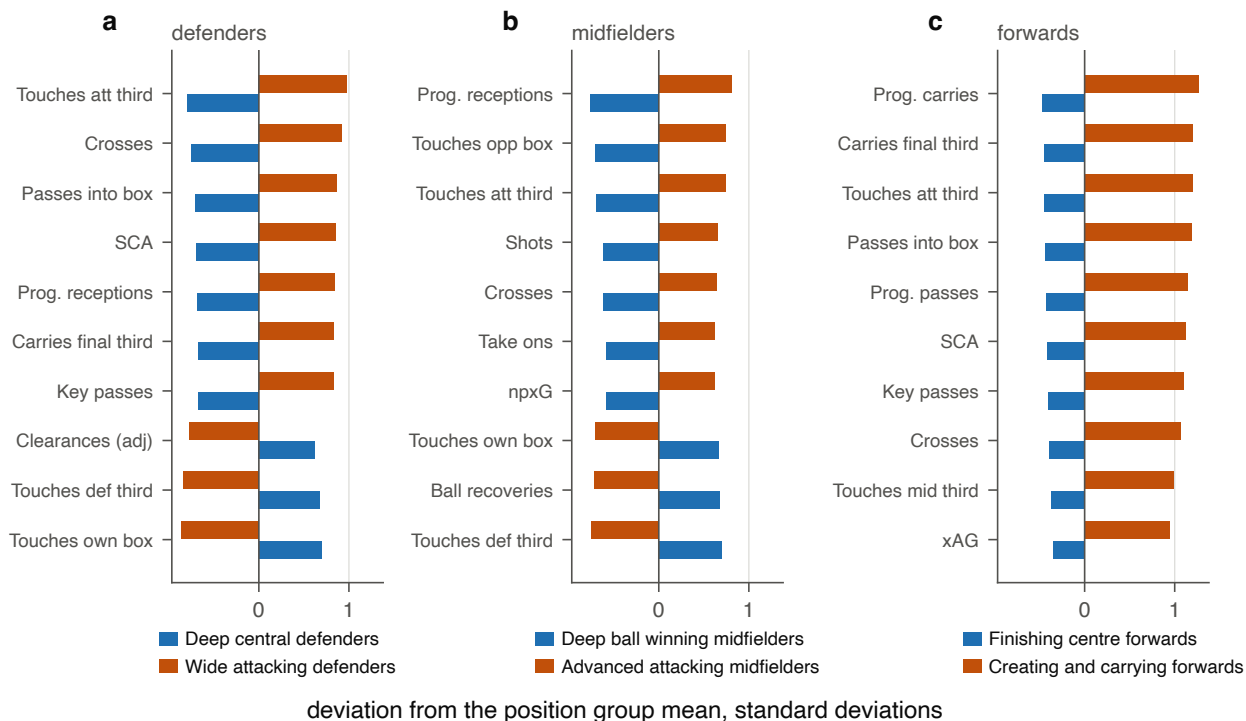


Figure 3: What separates the two modes inside each position group, as the standardised deviation of each archetype centroid from its group mean on the ten features with the largest separation. **a**, defenders. **b**, midfielders. **c**, forwards. The two profiles within a group are mirror images by construction, since features are standardised within group before the split, so each panel describes one axis and its two ends. Radar views of the same centroids are in Figure S7.

description treating attacking involvement as one axis would miss that, and it is the clearest case in the paper of the richer feature set buying something a poorer one could not.

Among forwards the division separates 1,241 Finishing centre forwards from 463 Creating and carrying forwards, and it is not a division by scoring, since non-penalty expected goals are slightly higher in the first group. What separates them is carrying and creation, with the second recording nearly three times the progressive carries, almost three times the passes into the box and twice the take-ons while shooting slightly less. Forward is not one role, but the axis that splits it is involvement in build-up rather than goal threat.

One structural property constrains how far the six read as independent. Features are standardised within position group before a two-way split, so the two centroids in a group are exactly anti-parallel and the larger cluster necessarily carries the smaller deviations. Each pair is the two ends of one axis, and the six archetypes describe three axes rather than six independently shaped profiles. The three are also not equally supported. The defender split is strongest on every measure applied to it, the midfield split is well supported with a more populated boundary, and the forward split is the weakest and should be read with the most caution, since its dip test does not reject unimodality, the division is uneven, and one league’s refit falls well below the others.

3.4 The same division in every season, league and feature family

A partition that appears once describes one dataset; a partition that appears independently in five seasons and five competitions describes the sport. Each season and each league is therefore

standardised inside itself and clustered from scratch, borrowing nothing from the pooled fit.

Every scope recovers the pooled partition. The weakest season agrees with the pool at an adjusted Rand index of 0.916 and the weakest league at 0.932, on 9,263 player-seasons, and the correlation between each scope’s own discriminant direction and the pooled direction never falls below 0.9983, so the separating axis is the same object each time rather than a similarly sized one. Importing one scope’s centroids into another scope’s players is the cleaner test, since it separates replication of the solution from change in the players: across the twenty ordered league pairs the transferred centroids reproduce the receiving league’s own fit at a mean adjusted Rand index of 0.944, never below 0.893. A model of role fitted on Ligue 1 assigns Bundesliga players almost exactly as one fitted on Bundesliga does. Season against season on players present in both is lower, at 0.693, and the pattern explains why: agreement decays monotonically with the gap between seasons while the transfer index over the same pairs stays flat, so the decay is players changing role over four years. Across those pairs 91.6 percent of players are assigned to the same mode in both seasons, so roughly one player in twelve moves between modes over a four-year span, a plausible rate of genuine role change rather than the churn an unstable partition would produce.

A result on 33 features could also be carried by one block of them, and a reviewer who suspected the modes were the expected goals block wearing a costume would be right to ask. Removing any one of the six families removes nothing that matters: the rule returns two clusters in every leave-one-out in every scope, 51 of the 52 feature sets tested exceed every one of the 25 clusterless simulations, and the adjusted Rand index between each ablated partition and the full one runs from 0.703 to 0.990, so the survivors are the same two modes rather than two different ones. The converse is more instructive than the headline. Most families produce a two-cluster solution that beats the null when used alone, but agreement with the full partition shows what those solutions are: shooting alone in defenders separates from the null at 19.79 standard deviations while agreeing with the paper’s partition at only 0.000, so it finds the finishers against everyone else, not the deep against the wide. Only territory reproduces the paper’s own modes by itself in every scope, at an adjusted Rand index from 0.732 to 0.856. The two modes are where a player touches the ball, and every other family either sharpens that division or, alone, cuts a different one. Figure 4 shows the refits above and the ablation below, and Table S5 gives every cell.

3.5 Do the modes predict results?

Everything above is a claim about the structure of the data. The archive also carries what happened, so the modes can be tested against real results. Each of 490 team-seasons is summarised by the minutes-weighted share of its outfield minutes in the attacking mode and by the mean position of its players along the axis, with eligible players covering a median 96.9 percent of each club’s outfield minutes.

The share correlates with points per match at a Spearman rank correlation of 0.430 (0.348 to 0.502), and the continuous axis position is stronger at 0.655, which is what a continuum should show: where a club’s players sit along the axis carries more than which side of the cut they fall on. Table 1 lists the ten team-seasons at the extremes so a reader can check them against memory, and they are the sides one would name.

A correlation between attacking composition and points is partly circular, because good teams have good attackers, so the test that matters is whether the modes carry anything beyond team possession. Ridge regression of points per match on possession alone, the composition alone and both gives out-of-sample coefficients of determination of 0.566, 0.545 and 0.664 holding out one season at a time, and 0.558, 0.546 and 0.661 under the strictest holdout, one club at a time, where no held-out team-season shares a club with the training data. The modes add 0.099 (0.067 to 0.131)

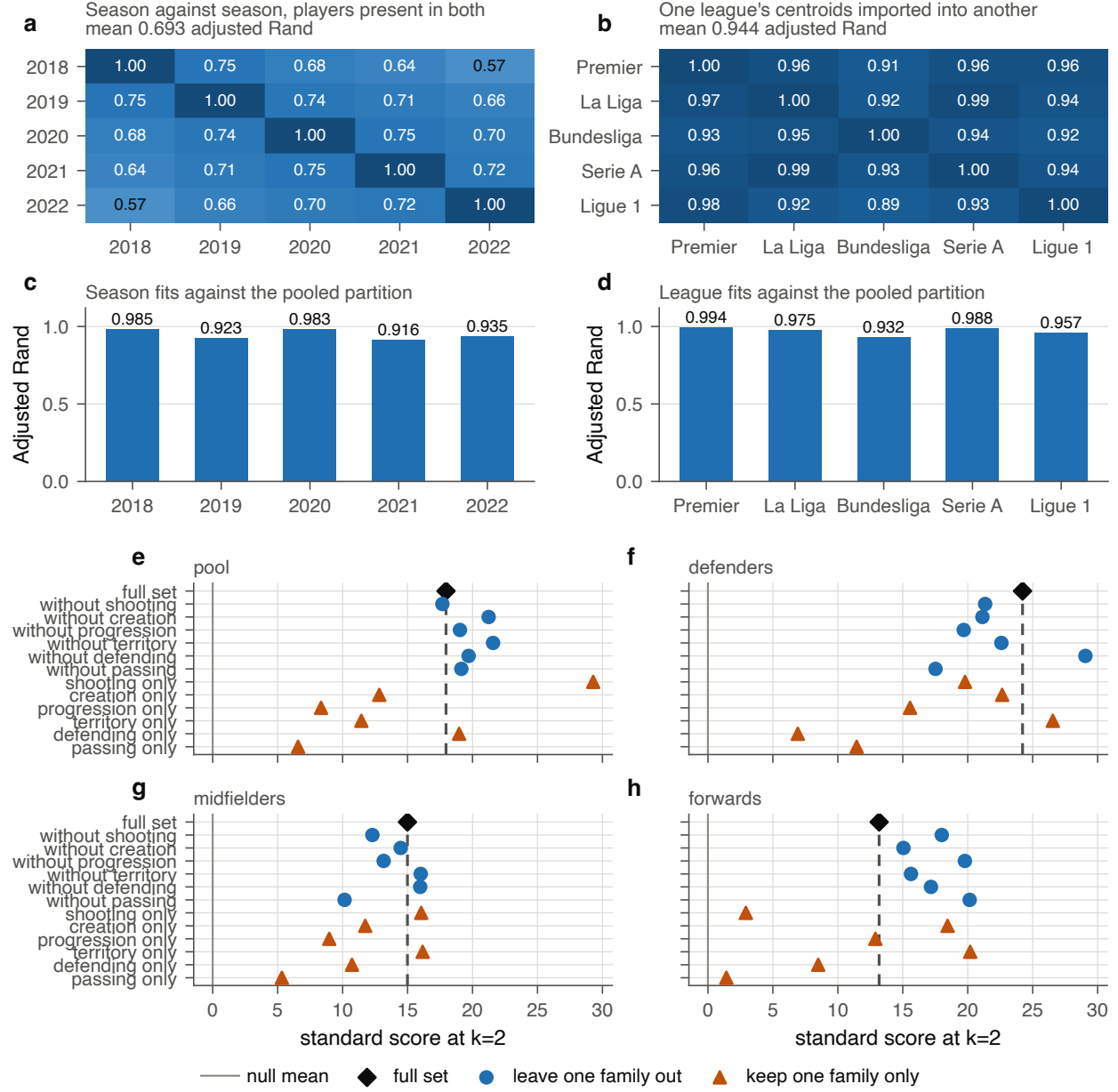


Figure 4: The division reproduces across every partition of the data and every part of the feature set. **a**, agreement between seasons on the players common to both, which declines as the gap widens because players change. **b**, one league's centroids imported into another league's players, which does not decline. **c** and **d**, each season and each league fitted from scratch and scored against the pooled partition. **e** to **h**, standard score of the observed two-cluster silhouette against the clusterless null under every feature-family ablation, one panel per scope: circles remove one family, triangles keep one family alone, and the vertical line is the full feature set.

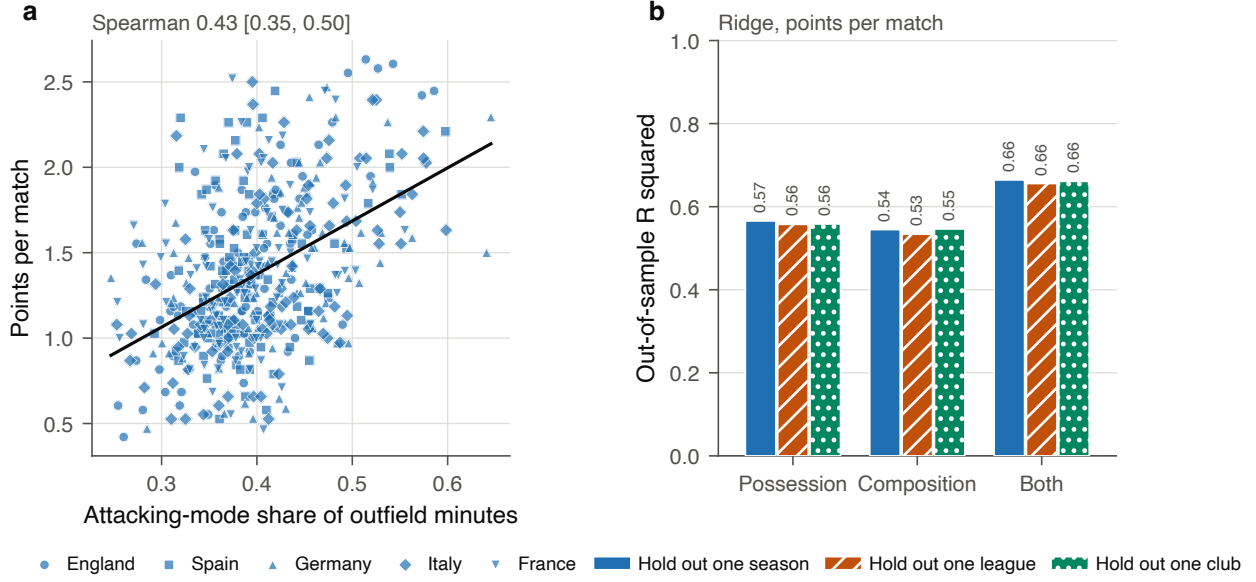


Figure 5: The two modes against real results over 490 team-seasons. **a**, the minutes-weighted share of outfield minutes in the attacking mode against points per match, one marker shape per league. **b**, out-of-sample coefficient of determination for points per match from possession alone, the composition alone and both, under leave-one-season-out, leave-one-league-out and leave-one-club-out holdouts.

to what possession explains under the first and 0.102 under the last. The composition alone predicts slightly worse than possession alone, which should be said plainly: the modes are not a substitute for dominance, they are information about how it is produced that dominance does not contain.

Where the information lives is the sharpest result here. Possession plus the two two-mode measures alone reach 0.660, an increment of 0.094 over possession that is the whole of the gain. Adding the six archetype shares on top adds 0.005 (-0.006 to 0.015), an interval that contains zero. Real results say what the null calibration said: two modes carry information and a finer taxonomy carries none the two modes did not already carry. Figure 5 shows the share against points and the out of sample comparison.

3.6 Goalkeepers

Goalkeepers are described by a different vocabulary, and the primary sample supplies all of it across 1,039 keeper-seasons. The first attempt at a partition walks into a trap, and the trap is itself a finding. Clustering on all available features returns 2 clusters that look interpretable until one asks what they are made of: membership explains a mean of 0.386 of the variance in situational quantities, the volume and quality of shooting a defence permits, against 0.109 in quantities describing what the goalkeeper does. The partition is largely a ranking of the teams in front of the goalkeepers. Named as archetypes it would be a league table wearing a tactical vocabulary. Clustering on technique alone reverses that completely: the same rule again returns 2 clusters, at a bootstrap adjusted Rand index of 0.570, and membership now explains 0.135 of the variance in technique against 0.019 in situation, a ratio of about seven.

What that partition describes is not the expected taxonomy. The familiar trio of shot stopper, sweeper keeper and distributor is not what the data produces. The division is between Line keepers,

Table 1: The five team-seasons with the highest and the five with the lowest minutes-weighted share of outfield minutes in the attacking mode, with their real points per match and final league position.

Club	League	Season	Share	Points per match	Position
<i>Highest attacking-mode share</i>					
Bayern Munich	Bundesliga	2019	0.645	2.29	1
Hoffenheim	Bundesliga	2019	0.641	1.50	9
Fiorentina	Serie A	2022	0.598	1.63	7
Real Madrid	La Liga	2021	0.598	2.21	2
Manchester City	Premier League	2022	0.586	2.45	1
<i>Lowest attacking-mode share</i>					
Mainz 05	Bundesliga	2022	0.247	1.35	8
Parma	Serie A	2019	0.253	1.08	16
Reims	Ligue 1	2022	0.254	1.21	12
Sheffield Utd	Premier League	2021	0.254	0.60	20
Troyes	Ligue 1	2022	0.256	1.00	15

411 keeper-seasons who stay on their line, and Sweeper claimers, 282 who come off it, separating on cross stopping and sweeping together, while distribution style and shot stopping ability barely separate the clusters at all and lie as continua across the division rather than as dimensions of it. The results test discriminates between the two partitions in the direction the effect sizes predict: the full-feature partition’s share of a club’s goalkeeper minutes correlates with points per match at 0.713, which is what a partition sorting keepers by the team in front of them must do, while the technique partition’s correlates at 0.087 (0.002 to 0.174), an interval barely clearing zero. The line keeper against sweeper claimer division is unrelated to how good the team is, which is the property a description of goalkeeping rather than of goalkeepers’ teams should have, and it echoes the outfield result: one binary distinction plus continuous variation, not a set of types. Figure 6 shows the technique partition and the two profiles.

3.7 Robustness

Four checks bound what the result depends on, each reported in full in the supplement.

It does not depend on the richness of the description. The reduced sample is an unusually clean test, because the withdrawal removed a specific and large part of the feature set without touching the population, and repeating the analysis there returns two clusters globally and within every position group, agreement with listed positions still weak at 0.226, and density-based clustering again assigning every position group entirely to noise. That also bounds what richer data buys: it sharpens the description considerably but reveals no additional modes the poorer one was hiding. It does not depend on the eligibility threshold either, since repeating the whole selection and stability procedure at 270, 450 and 900 minutes returns two clusters at every threshold with the silhouette moving only from 0.286 to 0.302 (Section S5).

The possession adjustment behind the defensive features is measured rather than assumed, and the measurement matters: four of the five defensive elasticities have bootstrap intervals excluding one, so the conventional proportional correction overcorrects them, and applying it reverses the sign of the residual correlation with team possession rather than removing it. The correction is also one sided, and that asymmetry is tested rather than asserted (Section S3). Finally the partitions are not artefacts of sampling noise in per-90 rates computed over partial seasons, since empirical Bayes shrinkage toward the position group mean leaves the global partition essentially unchanged while

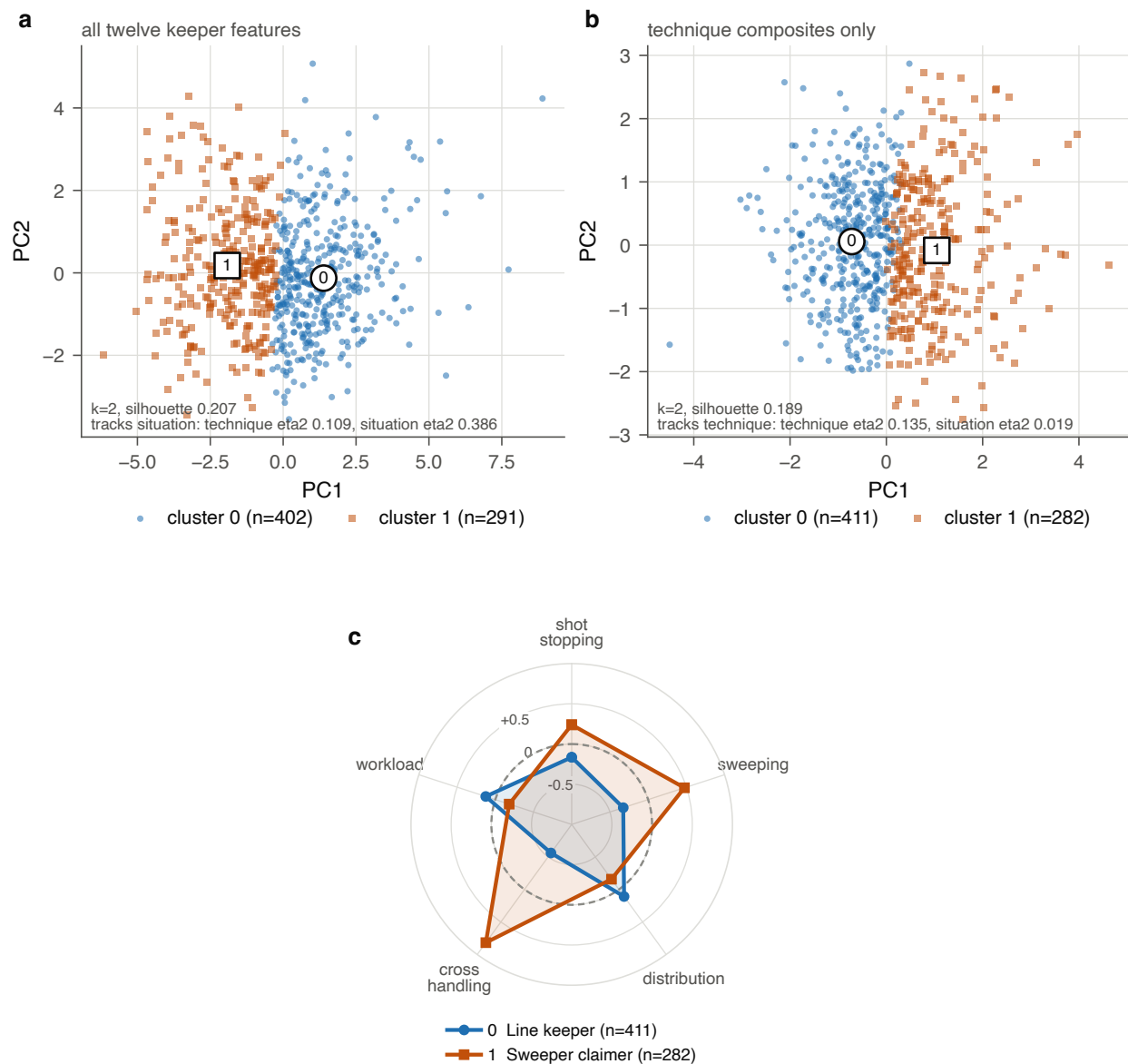


Figure 6: Goalkeeper structure on the primary sample. **a** and **b**, the technique partition, which separates goalkeepers who stay on their line from those who come off it. **c**, the two cluster centroids on the five composite axes, in standard deviations from the eligible pool mean, with the dashed ring marking that mean and the distribution axis signed so that a higher value means shorter passing. The clusters separate on cross handling and sweeping together, while distribution and shot stopping vary continuously across both rather than distinguishing them.

the within-group splits survive unequally; Section S10 reports that and the within-season role drift analysis that bounds the assumption that a player's role is constant across a season.

4 Discussion and conclusion

The result is easy to overstate in either direction. It does not say football has two kinds of player, and it does not say role is a single number. It says the distribution of players over 33 measured statistics has two regions of concentrated density joined by a populated middle, that the axis running between them is the dominant direction in the space, and that no division finer than that one is distinguishable from what a partitioning algorithm produces on data containing no divisions. That is a description a mixture representation expresses well and a taxonomy does not, which is why the archetypal and partial membership approaches to athlete description fit this data better than the genre that dominates it [8, 23].

The axis is territorial rather than technical. Its strongest loadings are touches by pitch third and progressive receptions, not passing accuracy or duel success, and the same ordering appears in the supervised attributions. Football’s own vocabulary mixes territory with technique freely, describing a player as a ball-playing centre back or a pressing forward, but the season aggregate record is far better at recording where a player was than what he did there. Any role structure recovered from it will be mostly a statement about territory, and reading it as a statement about ability is the most common way to misuse this kind of analysis. That the division cuts across the listed positions rather than reproducing them is consistent with what tracking and event data show about modern roles, where positional assignment is fluid within a match and formations are recovered as distributions rather than as slots [4, 5].

4.1 Why finer taxonomies keep being published

The methodological point generalises past football. A published role taxonomy is typically supported by three things: a selection criterion that picks the reported number of clusters, a stability check showing the partition survives resampling, and centroid profiles that read sensibly against domain knowledge. Every one of those is satisfied here by partitions this paper shows to be artefacts, because none of the three tests existence. A selection criterion compares candidate partitions against each other and is never offered the option of returning none, which is the limitation the stopping-rule literature has documented since it began [12, 13]. A bootstrap measures whether an algorithm applied twice to similar data behaves similarly, which a deterministic procedure on a smooth unimodal distribution does very reliably, exactly as the theory of stability-based validation predicts [15, 16]. Interpretability is the weakest of the three, because a cut through any continuous distribution produces two centroids that differ along the direction of the cut, and a domain expert can narrate that difference whether or not it corresponds to a real boundary. The evidence here makes the concern measured rather than hypothetical: on data built to contain no groups the same pipeline returns a silhouette maximum at two clusters, two nameable centroids, and a bootstrap adjusted Rand index of at least 0.911 in every scope, a difference from the real data as small as 0.003. Given that 55 percent of sports science clustering studies do not state how the number of clusters was chosen at all, the gap between the applied genre and the validity literature is wide [11].

Calibration against matched clusterless data closes it at low cost. It requires no assumption the analysis does not already make, it costs a few dozen refits, and it is the single step separating the two-mode result reported here from the six-way taxonomy the same pipeline would have produced with the same apparent confidence. The gap statistic already does a version of this against a uniform reference [14], and what this paper adds is not a new statistic but harder references and a reporting convention, that the observed value be placed against all of them so a reader can see which the claim survives. Hartigan’s dip test and a density-based method permitted to answer that nothing is clustered are useful supplements [18, 19, 32], and the agreement of all three is what makes the

negative half of the result credible. The position is the one Hennig argues for, that a partition should be judged by whether it has the properties the analyst needs rather than by whether it is correct [20]; the property asked for here is that a partition be distinguishable from one imposed on a continuum. The comparison with the closest prior work supports it. Akhanli and Hennig’s aggregated validity index, calibrated against random partitions of the observed data rather than against clusterless data, returned a coarse partition into five ordered groups running along the same territorial axis found here [21, 22]. Two independent calibrations agree that the recoverable structure in season aggregates is coarse and ordered, and that the taxonomies of a dozen named types the applied genre reports are not what either supports [1–3, 9].

4.2 What follows in practice

The division leaves the two extreme position groups intact and cuts the group between them close to in half, and the players it reassigns are recognisable: holding midfielders sit with the defenders and attacking midfielders with the forwards. That the supervised model errs in the same place is corroboration from machinery sharing nothing with the clustering. Listed position is therefore a reasonable but coarse summary at the extremes and close to arbitrary in the middle, which is what anyone who has watched a modern midfield would expect and which the data now support quantitatively.

The test against real outcomes closes the gap between describing the statistics and describing the sport, and it closes it in two directions at once. The two-mode composition of a club carries information about its points that possession does not, out of sample under three different holdouts, so the axis the partition cuts is one along which football is actually decided. The six archetypes carry no information about points the two modes did not already carry, so the finer taxonomy fails against results exactly as it failed against the clusterless null. Those are independent tests, one internal and one external, and they agree. Two cautions belong with that: the composition alone predicts slightly worse than possession alone, so the increment is the right thing to cite, and the within-league correlation varies from 0.19 in La Liga to 0.61 in Premier League.

For recruitment the honest use of this work is narrow and real. The involvement axis is stable enough to support questions about squad composition, such as whether a club’s minutes are concentrated at one end of the distribution. What it does not support is a scouting taxonomy, since a shortlist built by matching a target’s archetype label matches a label the data do not establish, and a similarity search over the continuous profile is the better instrument because it does not force a boundary that is not there. The two directly transferable items are the possession elasticities, which apply to any defensive count comparison and correct a convention that has stood since it was proposed without being measured [29], and the null calibration, which costs little and should be run before any archetype claim is made. The goalkeeper analysis carries its own caution: a partition built from the full public feature block is mostly a partition of the teams in front of the goalkeepers, so anyone building goalkeeper archetypes from public data will produce labels describing a role assignment made by a coach rather than a property of the goalkeeper, and conditioning on shot difficulty through post-shot expected goals is the minimum requirement for saying anything about shot stopping at all [36].

4.3 Limitations and future work

Season aggregates discard context. A tackle count does not say where the tackles happened, an expected goals total does not distinguish six poor chances from two good ones, and a touch map summarised into three thirds is a coarse instrument for a question that turns out to be territorial.

Everything here is a claim about what season aggregate data can support, not about what football contains, and the two should not be conflated: finer roles may well exist where this description cannot see them, and event and tracking data would be the way to find out [6, 7]. The confound possession adjustment does not remove is team strength, since attacking statistics still scale with a team’s own share of the ball. The unit of analysis is the player-season and player-seasons are nested, with the same player appearing in up to five rows and clubs nested within leagues, so pooled intervals are narrower than a fully nested analysis would give; the defence against this is design rather than adjustment, since every central claim is also made on independent refits by season and by league and the test against results holds out whole seasons, leagues and clubs. The archive is a mirror rather than the source, and one season was excluded outright because a whole feature block was absent from it. Season aggregation also assumes a player’s role is constant across a season, which holds for most players and fails for a minority.

The question this paper set out to answer is whether football player roles, as season aggregate statistics describe them, form types or a continuum. They form neither cleanly. The useful description sits between the two: an ordered space with one dominant axis, density concentrated at two ends, and a populated middle that no partition divides well. Two modes are real, they reproduce in every season and every competition, they predict what happened on the pitch, and nothing finer survives a test that structureless data would fail. What follows for practice is a caution about method rather than about football, because a silhouette maximum, a high bootstrap adjusted Rand index and an interpretable centroid are all obtainable from data containing no groups whatsoever: the algorithm supplies the groups.

Author contributions

C.C. conceived the research question. Z.C. designed the study, carried out the analysis, and wrote the paper. Both authors approved the final version.

Data and code availability

The analysis code, the retrieval step that fetches the archive from its stated upstream and records a digest per file, the cluster assignments for both samples and the sources of this document are available at <https://github.com/ZacharyTChung/pl-player-profiling>. The primary sample is the public worldfootballR mirror of FBref described in Section 2.1, and the reduced sample was retrieved from FBref and Understat as described there.

Acknowledgements

The statistics analysed here originate with FBref and were retrieved from the public mirror maintained by the worldfootballR project [26]. The analysis code and the manuscript were developed with the assistance of an AI coding assistant, Claude, under the direction of the authors, who take full responsibility for the content.

References

- [1] Soumaya El Foukahi, Kaoutar El Handri, Mohamed El Ghmary, and Hassan Echoukairi. Unveiling midfielder roles in elite football: A data-driven clustering approach. In *Artificial*

- Intelligence and Mathematics*, Springer Proceedings in Mathematics & Statistics, pages 465–477. Springer, 2026. doi: 10.1007/978-3-032-18042-1_37.
- [2] Samuel Kalman and Jonathan Bosch. NBA lineup analysis on clustered player tendencies: A new approach to the positions of basketball and modeling lineup efficiency of soft lineup aggregates. In *MIT Sloan Sports Analytics Conference*, 2020. URL <https://www.sloansportsconference.com/research-papers/nba-lineup-analysis-on-clustered-player-tendencies-a-new-approach-to-the-positions-of-basketball-modeling-lineup-efficiency>.
 - [3] Muthu Alagappan. From 5 to 13: Redefining the positions in basketball. In *MIT Sloan Sports Analytics Conference*, 2012. URL <https://www.sloansportsconference.com/research-papers/from-5-to-13-redefining-the-positions-in-basketball>.
 - [4] Alina Bialkowski, Patrick Lucey, Peter Carr, Yisong Yue, Sridha Sridharan, and Iain Matthews. Large-scale analysis of soccer matches using spatiotemporal tracking data. In *Proceedings of the 2014 IEEE International Conference on Data Mining (ICDM)*, pages 725–730, 2014. doi: 10.1109/ICDM.2014.133. URL <https://doi.org/10.1109/ICDM.2014.133>.
 - [5] Tom Decroos and Jesse Davis. Player vectors: Characterizing soccer players’ playing style from match event streams. In *Machine Learning and Knowledge Discovery in Databases: ECML PKDD 2019*, Lecture Notes in Computer Science, pages 569–584. Springer, 2020. doi: 10.1007/978-3-030-46133-1_34. URL https://doi.org/10.1007/978-3-030-46133-1_34.
 - [6] Luca Pappalardo, Paolo Cintia, Paolo Ferragina, Emanuele Massucco, Dino Pedreschi, and Fosca Giannotti. PlayeRank: Data-driven performance evaluation and player ranking in soccer via a machine learning approach. *ACM Transactions on Intelligent Systems and Technology*, 10(5), 2019. doi: 10.1145/3343172. URL <https://doi.org/10.1145/3343172>. arXiv:1802.04987.
 - [7] Bart Aalbers and Jan Van Haaren. Distinguishing between roles of football players in play-by-play match event data, 2018. URL <https://arxiv.org/abs/1809.05173>.
 - [8] Emiliano Seri, Roberto Rocci, and Thomas Brendan Murphy. Partial membership models for soft clustering of multivariate football player performance data, 2024. URL <https://arxiv.org/abs/2409.01874>.
 - [9] Spyridon Plakias, Serafeim Moustakidis, Christos Kokkotis, Marina Papalexi, Themistoklis Tsatalas, Giannis Giakas, and Dimitrios Tsaopoulos. Identifying soccer players’ playing styles: A systematic review. *Journal of Functional Morphology and Kinesiology*, 8(3):104, 2023. doi: 10.3390/jfmk8030104.
 - [10] Linda Gu, Wenlong Wang, Spyridon Plakias, and Shaoliang Zhang. Playing style identification in team sports: A systematic review from statistical dimensionality reduction to unsupervised machine learning. *International Journal of Sports Science & Coaching*, 20(6):2742–2761, 2025. doi: 10.1177/17479541251372586.
 - [11] Daniel Fernández, Martí Casals, Martí Oliver, Montse Plensa, and Marica Manisera. Reporting of clustering techniques in sports sciences: a scoping review. *Electronic Journal of Applied Statistical Analysis*, 17(3):653–675, 2024. doi: 10.1285/i20705948v17n3p653.
 - [12] Glenn W. Milligan and Martha C. Cooper. An examination of procedures for determining the number of clusters in a data set. *Psychometrika*, 50(2):159–179, 1985. doi: 10.1007/BF02294245.

- [13] Peter J. Rousseeuw. Silhouettes: A graphical aid to the interpretation and validation of cluster analysis. *Journal of Computational and Applied Mathematics*, 20:53–65, 1987. doi: 10.1016/0377-0427(87)90125-7. URL [https://doi.org/10.1016/0377-0427\(87\)90125-7](https://doi.org/10.1016/0377-0427(87)90125-7).
- [14] Robert Tibshirani, Guenther Walther, and Trevor Hastie. Estimating the number of clusters in a data set via the gap statistic. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 63(2):411–423, 2001. doi: 10.1111/1467-9868.00293. URL <https://doi.org/10.1111/1467-9868.00293>.
- [15] Asa Ben-Hur, André Elisseeff, and Isabelle Guyon. A stability based method for discovering structure in clustered data. In *Pacific Symposium on Biocomputing*, volume 7, pages 6–17, 2002. URL <https://psb.stanford.edu/psb-online/proceedings/psb02/benhur.pdf>.
- [16] Ulrike von Luxburg. Clustering stability: An overview. *Foundations and Trends in Machine Learning*, 2(3):235–274, 2010. doi: 10.1561/2200000008.
- [17] Margareta Ackerman and Shai Ben-David. Clusterability: A theoretical study. In *Proceedings of the Twelfth International Conference on Artificial Intelligence and Statistics (AISTATS 2009)*, volume 5 of *Proceedings of Machine Learning Research*, pages 1–8, 2009. URL <https://proceedings.mlr.press/v5/ackerman09a.html>.
- [18] Andreas Adolffson, Margareta Ackerman, and Naomi C. Brownstein. To cluster, or not to cluster: An analysis of clusterability methods. *Pattern Recognition*, 88:13–26, 2019. doi: 10.1016/j.patcog.2018.10.026.
- [19] J. A. Hartigan and P. M. Hartigan. The dip test of unimodality. *The Annals of Statistics*, 13(1):70–84, 1985. doi: 10.1214/aos/1176346577.
- [20] Christian Hennig. What are the true clusters? *Pattern Recognition Letters*, 64:53–62, 2015. doi: 10.1016/j.patrec.2015.04.009.
- [21] Serhat Emre Akhanli and Christian Hennig. Comparing clusterings and numbers of clusters by aggregation of calibrated clustering validity indexes. *Statistics and Computing*, 30(5):1523–1544, 2020. doi: 10.1007/s11222-020-09958-2.
- [22] Serhat Emre Akhanli and Christian Hennig. Clustering of football players based on performance data and aggregated clustering validity indexes. *Journal of Quantitative Analysis in Sports*, 19(2):103–123, 2023. doi: 10.1515/jqas-2022-0037.
- [23] Manuel J. A. Eugster. Performance profiles based on archetypal athletes. *International Journal of Performance Analysis in Sport*, 12(1):166–187, 2012. doi: 10.1080/24748668.2012.11868592.
- [24] Fabian Pedregosa, Gaël Varoquaux, Alexandre Gramfort, Vincent Michel, Bertrand Thirion, Olivier Grisel, Mathieu Blondel, Peter Prettenhofer, Ron Weiss, Vincent Dubourg, Jake Vanderplas, Alexandre Passos, David Cournapeau, Matthieu Brucher, Matthieu Perrot, and Édouard Duchesnay. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830, 2011. URL <https://jmlr.org/papers/volume12/pedregosa11a/pedregosa11a.pdf>.
- [25] Sports Reference LLC. FBref.com: Football statistics and history. <https://fbref.com/en/>, 2026. Advanced statistics were licensed from a commercial event-data provider, StatsBomb until 2022 and Opta thereafter; the provider-derived columns were withdrawn on 20 January 2026.

- [26] Jason Zivkovic. worldfootballR: Extract and clean world football (soccer) data. R package and public data mirror, 2026. URL https://github.com/JaseZiv/worldfootballR_data. Archived FBref advanced season tables retrieved 7 September 2026.
- [27] Harm Eggels, Ruud van Elk, and Mykola Pechenizkiy. Explaining soccer match outcomes with goal scoring opportunities predictive analytics. In *Proceedings of the 3rd Workshop on Machine Learning and Data Mining for Sports Analytics (MLSA 2016), co-located with ECML/PKDD 2016*, volume 1842 of *CEUR Workshop Proceedings*, 2016. URL https://ceur-ws.org/Vol-1842/paper_07.pdf.
- [28] Alex Rathke. An examination of expected goals and shot efficiency in soccer. *Journal of Human Sport and Exercise*, 12(Proc2), 2017. doi: 10.14198/jhse.2017.12.proc2.05. URL <https://doi.org/10.14198/jhse.2017.12.proc2.05>.
- [29] Ted Knutson. Introducing possession-adjusted player stats. StatsBomb, June 2014. URL <https://blogarchive.statsbomb.com/articles/soccer/introducing-possession-adjusted-player-stats/>. Published 23 June 2014.
- [30] Pieter Robberechts. soccerdata: A collection of scrapers for gathering soccer data. Python package, version 1.9.1, 2026. URL <https://github.com/probberechts/soccerdata>.
- [31] Understat. Understat: Expected goals statistics. <https://understat.com>, 2026. Player and team expected goals for the seasons ending 2025 and 2026, retrieved 2026.
- [32] Ricardo J. G. B. Campello, Davoud Moulavi, and Jörg Sander. Density-based clustering based on hierarchical density estimates. In *Advances in Knowledge Discovery and Data Mining (PAKDD 2013)*, volume 7819 of *Lecture Notes in Computer Science*, pages 160–172. Springer, 2013. doi: 10.1007/978-3-642-37456-2_14. URL https://doi.org/10.1007/978-3-642-37456-2_14.
- [33] Lawrence Hubert and Phipps Arabie. Comparing partitions. *Journal of Classification*, 2:193–218, 1985. doi: 10.1007/BF01908075. URL <https://doi.org/10.1007/BF01908075>.
- [34] Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu. LightGBM: A highly efficient gradient boosting decision tree. In *Advances in Neural Information Processing Systems 30 (NeurIPS 2017)*, 2017. URL <https://proceedings.neurips.cc/paper/2017/hash/6449f44a102fde848669bdd9eb6b76fa-Abstract.html>.
- [35] Scott M. Lundberg and Su-In Lee. A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems 30 (NeurIPS 2017)*, 2017. URL https://proceedings.neurips.cc/paper_files/paper/2017/hash/8a20a8621978632d76c43dfd28b67767-Abstract.html. arXiv:1705.07874.
- [36] Dinesh Vatvani. Upgrading Expected Goals. StatsBomb Blog, 2022. URL <https://blogarchive.statsbomb.com/articles/soccer/upgrading-expected-goals/>.